

CSC3209 Software architecture and Design Patterns

Lecture8: Architecture and Requirements

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References

- Materials from
 - Chapter 16: Len Bass, Paul Clements, Rick Kazman, Software Architecture in Practice, 3/E

Lecture Outline

- Gathering ASRs from Requirements Documents
- Gathering ASRs by Interviewing Stakeholders
- Gathering ASRs by Understanding the Business Goals
- Capturing ASRs in a Utility Tree
- Tying the Methods Together
- Summary

Requirements

- Architectures exist to build systems that satisfy requirements.
- But, to an architect, not all requirements are created equal.
- An *architecturally significant requirement* (ASR) is a requirement that will have a profound effect on the architecture.
- How do we find those?

ASRs and Requirements Documents

- An obvious location to look for candidate ASRs is in the requirements documents or in user stories.
- Requirements should be in requirements documents!
- Unfortunately, this is not usually the case.
- Why?

Don't Get Your Hopes Up

- Many projects don't create or maintain the detailed, high-quality requirements documents.
- Standard requirements pay more attention to functionality than quality attributes.
- Most of what is in a requirements specification does not affect the architecture.
- No architect just sits and waits until the requirements are “finished” before starting work. The architect *must* begin while the requirements are still in flux.

Don't Get Your Hopes Up

- **Quality** attributes, **when captured** at all, are often **captured poorly**.
 - “The system shall be modular”
 - “The system shall exhibit high usability”
 - “The system shall meet users’ performance expectations”
- Much of what is useful to an architect is not in even the best requirements document.
 - ASRs often derive from business goals in the development organization itself
 - Developmental qualities (such as teaming) are also out of scope

Sniffing Out ASRs

- Although requirements documents won't tell an architect the whole story, they are an important source of ASRs.
- Of course, ASRs aren't going to be conveniently labeled as such; the architect is going to have to perform a bit of excavation and archaeology to ferret them out.
- Lecture 5 categorizes the design decisions that architects have to make.
- This table summarizes each category of architectural design decision, and it gives a list of requirements to look for that might affect that kind of decision.
- If a requirement affects the making of a critical architectural design decision, it is by definition an ASR.

Design Decision Category	Look for Requirements Addressing . . .
Allocation of Responsibilities	Planned evolution of responsibilities, user roles, system modes, major processing steps, commercial packages
Coordination Model	Properties of the coordination (timeliness, currency, completeness, correctness, and consistency) Names of external elements, protocols, sensors or actuators (devices), middleware, network configurations (including their security properties) Evolution requirements on the list above
Data Model	Processing steps, information flows, major domain entities, access rights, persistence, evolution requirements
Management of Resources	Time, concurrency, memory footprint, scheduling, multiple users, multiple activities, devices, energy usage, soft resources (buffers, queues, etc.) Scalability requirements on the list above
Mapping among Architectural Elements	Plans for teaming, processors, families of processors, evolution of processors, network configurations
Binding Time Decisions	Extension of or flexibility of functionality, regional distinctions, language distinctions, portability, calibrations, configurations
Choice of Technology	Named technologies, changes to technologies (planned and unplanned)

Gathering ASRs from Stakeholders

- Say your project won't have the QAs nailed down by the time you need to start your design work.
- What do you do?
- Stakeholders often have *no idea* what QAs they want in a system
 - if you insist on quantitative QA requirements, you're likely to get numbers that are arbitrary.
 - at least some of those requirements will be very difficult to satisfy.
- **Architects** often have very **good ideas** about **what QAs** are **reasonable** to provide.
- **Interviewing** the **relevant stakeholders** is the **surest way** to learn what they know and **need**.

Gathering ASRs from Stakeholders

- The results of stakeholder interviews should include
 - a list of architectural drivers
 - a set of QA scenarios that the stakeholders (as a group) prioritized.
- This information can be used to:
 - refine system and software requirements
 - understand and clarify the system's architectural drivers
 - provide rationale for why the architect subsequently made certain design decisions
 - guide the development of prototypes and simulations
 - influence the order in which the architecture is developed.

Quality Attribute Workshop

- The **QAW** is a facilitated, stakeholder-focused method to generate, prioritize, and refine quality attribute scenarios before the software architecture is completed.
- The **QAW** is focused on system-level concerns and specifically the role that software will play in the system.

QAW Steps

- **Step 1: QAW Presentation and Introductions.**

- QAW facilitators describe the motivation for the QAW and explain each step of the method.

- **Step 2: Business/Mission Presentation.**

- The stakeholder representing the business concerns behind the system presents the system's business context, broad functional requirements, constraints, and known quality attribute requirements.
- The quality attributes that will be refined in later steps will be derived largely from the business/mission needs presented in this step.

QAW Steps

- **Step 3: Architectural Plan Presentation.**

- The architect will present the system architectural plans as they stand.
- This lets stakeholders know the current architectural thinking, to the extent that it exists.

- **Step 4: Identification of Architectural Drivers.**

- The facilitators will share their list of key architectural drivers that they assembled during Steps 2 and 3, and ask the stakeholders for clarifications, additions, deletions, and corrections.
- The idea is to reach a consensus on a distilled list of architectural drivers that includes overall requirements, business drivers, constraints, and quality attributes.

QAW Steps

- **Step 5: Scenario Brainstorming.**

- Each stakeholder expresses a scenario representing his or her concerns with respect to the system.
- Facilitators ensure that each scenario has an explicit stimulus and response.
- The facilitators ensure that at least one representative scenario exists for each architectural driver listed in Step 4.

- **Step 6: Scenario Consolidation.**

- Similar scenarios are consolidated where reasonable.
- Consolidation helps to prevent votes from being spread across several scenarios that are expressing the same concern.

QAW Steps

- **Step 7: Scenario Prioritization.**

- **Prioritization** of the **scenarios** is accomplished by allocating each stakeholder a number of votes equal to 30 percent of the total number of scenarios

- **Step 8: Scenario Refinement.**

- The **top scenarios** are **refined** and **elaborated**.
- **Facilitators help** the **stakeholders put** the **scenarios** in the six-part scenario **form** of **source-stimulus-artifact-environment-response-response measure**.

Gathering ASRs by Understanding the Business Goals

- Common business goals include making a profit, improve capability/quality of system..
- Business goals are of interest to architects because they often are the precursor of requirements that may or may not be captured in a requirements specification but whose achievement (or lack) signals a successful (or less than successful) architectural design.
- Business goals frequently lead directly to ASRs.

Gathering ASRs by Understanding the Business Goals

- There are three possible **relationships between business goals** and an **architecture**:
 - **Business goals** often **lead** to **quality attribute requirements**
 - **Business goals** may **directly affect** the **architecture** without precipitating a **quality attribute** requirement at all
 - **No influence** at all.

Figure 16.1 illustrates the major points just described. In the figure, the **arrows** mean “**leads to**.” The **solid arrows** are the ones highlighting relationships of **most interest** to **architects**.

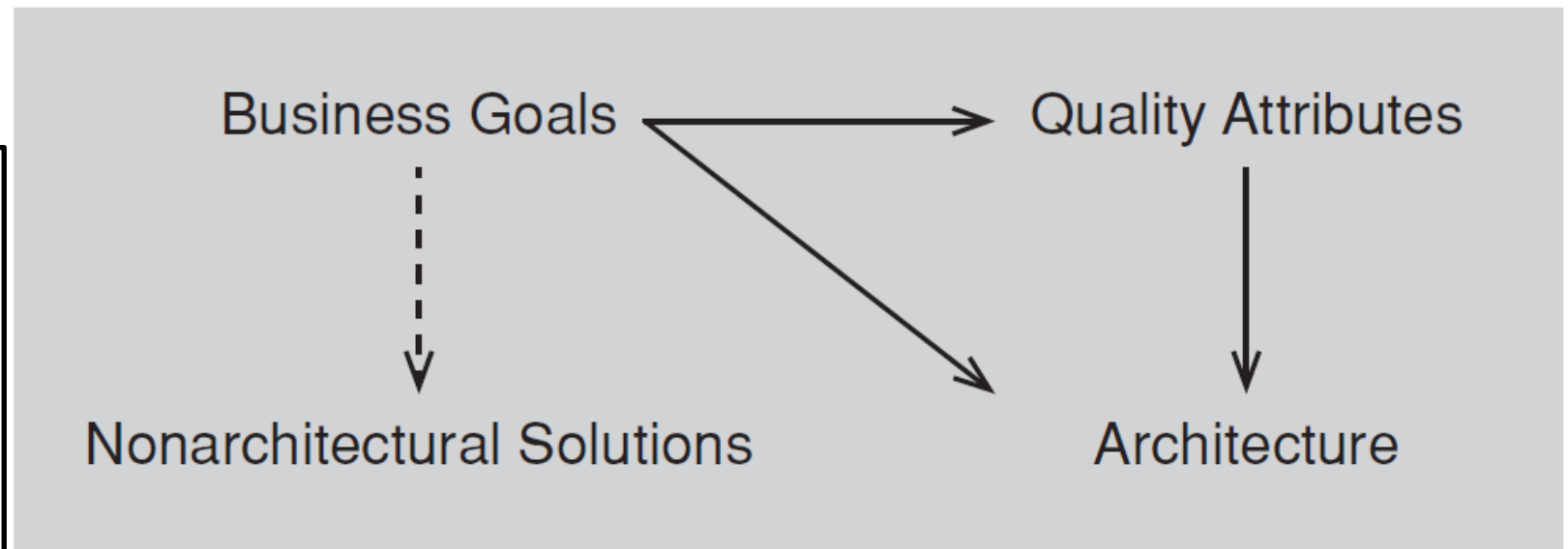


FIGURE 16.1 Some business goals may lead to quality attribute requirements (which lead to architectures), or lead directly to architectural decisions, or lead to nonarchitectural solutions.

Gathering ASRs by Understanding the Business Goals

- There are three possible relationships between business goals and an architecture:
 - Business goals often lead to quality attribute requirements.
 - every quality attribute requirement—such as user-visible response time or platform flexibility or ironclad security or any of a dozen other needs—should originate from some higher purpose that can be described in terms of added value.
 - for example, “Why do you want this system to have a really fast response time?”, we might hear that this will differentiate the product from its competition and let the developing organization capture market share

Gathering ASRs by Understanding the Business Goals

- There are three possible relationships between business goals and an architecture:
 - Business goals may directly affect the architecture without precipitating a quality attribute requirement at all
 - For example, an architect who designed a system without a database until the manager informed him that the database team needed work.
 - The architecture was importantly affected without any relevant quality attribute requirement.

Gathering ASRs by Understanding the Business Goals

- There are three possible relationships between business goals and an architecture:
 - No influence at all.
 - Not all business goals lead to quality attributes.
 - For example, a business goal to “reduce cost” may be realized by lowering the facility’s thermostats in the winter or reducing employees’ salaries or pensions.

Categories of Business Goals, to Aid in Elicitation

- Business goals are worth capturing explicitly. This is because they often imply ASRs that would otherwise go undetected until it is too late or too expensive to address them.
- Capturing business goals is well served by having a set of candidate business goals handy to use as conversation starters.
- Research in business goals has led us to adopt the categories. These categories can be used as an aid to brainstorming and elicitation.
- By employing the list of categories, and asking the stakeholders about possible business goals in each category, some assurance of coverage is gained.

Categories of Business Goals, to Aid in Elicitation

- These categories are not completely orthogonal. Some business goals may fit into more than one category, and that's all right.
- In an elicitation method, the categories should prompt questions about the existence of organizational business goals that fall into that category.

1.	Contributing to the growth and continuity of the organization
2.	Meeting financial objectives
3.	Meeting personal objectives
4.	Meeting responsibility to employees
5.	Meeting responsibility to society
6.	Meeting responsibility to state
7.	Meeting responsibility to shareholders
8.	Managing market position
9.	Improving business processes
10.	Managing the quality and reputation of products
11.	Managing change in environmental factors

Categories of Business Goals, to Aid in Elicitation

- **Contributing to the growth and continuity of the organization.**
- How does the system being developed contribute to the growth and continuity of the organization?
- In one experience using this business goal category, the system being developed was the sole reason for the existence of the organization.
 - If the system was not successful, the organization would cease to exist.
- Other topics that might come up in this category deal with market share, product lines, and international sales.

Categories of Business Goals, to Aid in Elicitation

- **Meeting financial objectives.**
- This category includes revenue generated or saved by the system.
- The system may be for sale, either in standalone form or by providing a service, in which case it generates revenue.
- The system may be for use in an internal process, in which case it should make those processes more effective or more efficient.
- Also in this category is the cost of development, deployment, and operation of the system.
- this category can also include financial objectives of individuals: a manager hoping for a raise, for example, or a shareholder expecting a dividend.

Categories of Business Goals, to Aid in Elicitation

- **Meeting personal objectives**
- Individuals have various goals associated with the construction of a system.
- They may range from “I want to enhance my reputation by the success of this system” to “I want to learn new technologies” to “I want to gain experience with a different portion of the development process than in the past.”
- In any case, it is possible that technical decisions are influenced by personal objectives.

Categories of Business Goals, to Aid in Elicitation

- **Meeting responsibility to employees**

- In this category, the employees in question are usually those employees involved in development or those involved in operation.
- Responsibility to employees involved in development might include ensuring that certain types of employees have a role in the development of this system, or it might include providing employees the opportunities to learn new skills.
- Responsibility to employees involved in operating the system might include safety, workload, or skill considerations.

Categories of Business Goals, to Aid in Elicitation

- **Meeting responsibility to society**
- Some organizations see themselves as being in business to serve society. For these organizations, the system under development is helping them meet those responsibilities.
- But all organizations must discharge a responsibility to society by obeying relevant laws and regulations.
- Other topics that might come up under this category are resource usage, “green computing,” ethics, safety, open source issues, security, and privacy.

Categories of Business Goals, to Aid in Elicitation

- **Meeting responsibility to state**
- **Government systems**, almost by definition, are intended to **meet responsibility** to a **state** or **country**.
- Other **topics** that might come up **in this category** deal with **export controls**, regulatory **conformance**, or supporting **government initiatives**.

Categories of Business Goals, to Aid in Elicitation

- **Meeting responsibility to shareholders**
- There is **overlap between this** category and the **financial objectives category**,
- but **additional topics** that might come up here are **liability protection** and **certain** types of **regulatory conformance** such as, in the United States, **adherence to the Sarbanes-Oxley Act**.

Categories of Business Goals, to Aid in Elicitation

- **Managing market position**

- Topics that might come up in this category are the strategy used to increase or hold market share, various types of intellectual property protection, or the time to market.

- **Improving business processes**

- Although this category partially overlaps with meeting financial objectives, reasons other than cost reduction exist for improving business processes. It may be that improved business processes enable new markets, new products, or better customer support.

Categories of Business Goals, to Aid in Elicitation

- **Managing the quality and reputation of products**

- Topics that might come up in this category include branding, recalls, types of potential users, quality of existing products, and testing support and strategies.

- **Managing change in environmental factors**

- The business context for a system might change. This item is intended to encourage the stakeholders to consider what might change in the business goals for a system.

Expressing Business Goals

- How will you write down a business goal once you've learned it?
- Just as for quality attributes, a scenario makes a convenient, uniform, and clarifying way to express business goals.
- A scenario helps ensure that all business goals are expressed clearly, in a consistent fashion, and contain sufficient information to enable their shared understanding by relevant stakeholders.

Expressing Business Goals

Business goal scenario, 7 parts

- *Goal-source*
- *Goal-subject*
- *Goal-object*
- *Environment*
- *Goal*
- *Goal-measure*
- *Pedigree and value*

Expressing Business Goals

Business goal scenario, 7 parts

- **Goal-source**

- The people or written artifacts providing the goal.

- **Goal-subject**

- The **stakeholders** who **own** the **goal** and **wish** it to be **true**.
- Each stakeholder might be an **individual** or the **organization** itself
- If the business **goal** is, for example, “**Maximize dividends for the shareholders,**” who is it that **cares** about that? It is probably **not** the **programmers** or the system’s **end users** (**unless** they happen to **own stock**).
- **Goal-subjects** can and do belong to **different organizations**. The **developing organization**, the **customer organizations**, **subcontractors**, **vendors** and **suppliers**, **standards bodies**, regulatory agencies, and organizations responsible or systems with which ours must interact are all potential goal-subjects.

Expressing Business Goals

Business goal scenario, 7 parts

- **Goal-object**

- The entities to which the goal applies.
- All goals have goal-objects: we want something to be true about something (or someone) that (or whom) we care about.
- For some goals the goal-object is clearly the development organization, but for some goals the goal-object can be more refined, such as the rank-and-file employees of the organization or the shareholders of the organization

- **Environment**

- The context for this goal
- Environment may be social, legal, competitive, customer, and technological
- Sometimes the political environment is key; this is as a kind of social factor. Upcoming technology may be a major factor.

Expressing Business Goals

Business goal scenario, 7 parts

- **Goal**
 - Any **business goal** articulated by the goal-source.
- **Goal-measure**
 - A **testable measurement** to **determine how** one would **know** if the **goal** has been **achieved**.
 - The goal-measure **should** usually **include** a **time component**, stating the **time by which** the goal should be **achieved**.
- **Pedigree and value**
 - The **degree** of **confidence** the person who stated the goal has in it
 - The **goal's volatility** and **value**.
 - The **value** of a goal **can be expressed** by **how much** its **owner** is **willing** to **spend** to achieve it or its relative importance compared to other goals.
 - **Relative importance** may be given by a **ranking** from **1 (most important)** to **n (least important)**, or by assigning each goal a value on a fixed scale such as 1 to 10 or high-medium-low.

Expressing Business Goals

- Elements 2–6 can be combined into a sentence that reads:

For the **system** being developed, **<goal-subject>** desires that **<goal object>** achieve **<goal>** in the context of **<environment>** and will be satisfied if **<goal-measure>**

- The sentence can be augmented by the goal's source (element 1) and the goal's pedigree and value (element 7).
- Example:

For **MySys**, the **project manager** has the goal that **customer satisfaction** will **rise by 10** percent (as a **result of** the **increased quality** of MySys).

A General Scenario for Business Goals

- A general scenario is a template for constructing specific or “concrete” scenarios. It uses the generic structure of a scenario to supply a list of possible values for each part of a scenario.
- See Table 16.4 for a general scenario for business goals.

TABLE 16.4 General Scenario Generation Table for Business Goals

2. Goal-subject	...	3. Goal-object	...	5. Goal	...	4. Environment	...	6. Goal-measure (examples, based on goal categories)	7. Value
Any stakeholder or stakeholder group identified as having a <i>legitimate</i> interest in the system	...	Individual	...	Contributing to the growth and continuity of the organization	...	Social (includes political)	...	Time that business remains viable	1– <i>n</i>
	has the goal that	System	...	Meeting financial objectives	...	Legal	...	Financial performance vs. objectives	1–10
	...	Portfolio	...	Meeting personal objectives	...	Competitive	...	Promotion or raise achieved in period	H-M-L
	...	Organization's employees	...	Meeting responsibility to employees	...	Customer	...	Employee satisfaction; turnover rate	Resources willing to expend
	...	Organization's shareholders	...	Meeting responsibility to society	...	Technological	...	Contribution to trade deficit/surplus	
	...	Organization	...	Meeting responsibility to state	...		...	Stock price, dividends	
	...	Nation	...	Meeting responsibility to shareholders	...		...	Market share	
	...	Society	...	Managing market position	...		...	Time to carry out a business process	
	...		...	Improving business processes	...		...	Quality measures of products	
	...		...	Managing quality and reputation of products	...		...	Technology-related problems	
	...		...	Managing change in environmental factors	...		...	Time window for achievement	

For each of these scenarios you **might** want to **additionally capture** its **source** (e.g., **Did this come directly from the goal-subject**, a **document**, a **third party**, a legal requirement?), its volatility, and its importance.

Capturing Business Goals

- One method for eliciting and documenting business goals is the Pedigreed Attribute eLicitation Method, or PALM.
- The word “pedigree” means that the business goal has a clear derivation or background.
- PALM uses the standard list of business goals and the business goal scenario format we described earlier.
- PALM can be used to sniff out missing requirements early in the life cycle.
 - For example, having stakeholders subscribe to the business goal of improving the quality and reputation of their products may very well lead to (for example) security, availability, and performance requirements that otherwise might not have been considered.

Capturing Business Goals

- PALM can also be used to discover and carry along additional information about existing requirements.
 - For example, a business goal might be to produce a product that outcompetes a rival's market entry.
 - This might precipitate a performance requirement for, say, half-second turnaround when the rival features one-second turnaround.
 - But if the competitor releases a new product with half-second turnaround, then what does our requirement become?
 - A conventional requirements document will continue to carry the half-second requirement, but the goal-savvy architect will know that the real requirement is to beat the competitor, which may mean even faster performance is needed.

Capturing Business Goals

- PALM can be used to examine particularly difficult quality attribute requirements to see if they can be relaxed.
 - We know of more than one system where a quality attribute requirement proved quite expensive to provide, and only after great effort, money, and time were expended trying to meet it was it revealed that the requirement had no actual basis other than being someone's best guess or fond wish at the time.

PALM: A Method for Eliciting Business Goals

- PALM is a **seven-step method**.
- Nominally **carried out** over **a day and a half** in a **workshop**.
- **Attended** by **architect(s)** and **stakeholders** who can speak to the relevant **business goals**.

PALM Steps

- ***PALM overview presentation***

- Overview of PALM, the problem it solves, its steps, and its expected outcomes.

- ***Business drivers presentation.***

- Briefing of business drivers by project management
- What are the goals of the customer organization for this system?
- What are the goals of the development organization?

- ***Architecture drivers presentation***

- Briefing by the architect on the driving business and quality attribute requirements: the ASRs.

PALM Steps

- ***Business goals elicitation***

- Using the standard business goal categories to guide discussion, we capture the set of important business goals for this system.
- Business goals are elaborated and expressed as scenarios.
- Consolidate almost-alike business goals to eliminate duplication.
- Participants prioritize the resulting set to identify the most important goals.

- ***Identification of potential quality attributes from business goals.***

- For each important business goal scenario, participants describe a quality attribute that (if architected into the system) would help achieve it. If the QA is not already a requirement, this is recorded as a finding.

PALM Steps

- *Assignment of pedigree to existing quality attribute drivers.*
 - For each architectural driver, named in step 3, identify which business goals it is there to support.
 - If none, that's recorded as a finding.
 - Otherwise, we establish its pedigree by asking for the source of the quantitative part.
 - For example: Why is there a 40-millisecond performance requirement? Why not 60 milliseconds? Or 80 milliseconds?
- *Exercise conclusion*
 - Review of results, next steps, and participant feedback.

Capturing ASRs in a Utility Tree

An **ASR** must have the following characteristics:

- A *profound impact on the architecture*
 - Including this requirement will very likely result in a different architecture than if it were not included.
- A *high business or mission value*
 - If the architecture is going to satisfy this requirement it must be of high value to important stakeholders.
- It is **helpful** to **record** them in **one place** so that the **list** can be **reviewed**, **referenced**, **used** to justify design decisions, and **revisited** over time or in the case of major system changes.
- Using a **single list** can also **help evaluate** each **potential ASR** against these **criteria**, and to **make sure** that **no architectural drivers**, **stakeholder** classes, or **business goals** are **lacking ASRs** that **express** their **needs**.

Utility Tree

- Architects can use a construct called a utility tree for all of these purposes.
- A utility tree begins with the word “utility” as the root node.
- Utility is an expression of the overall “goodness” of the system.
- We then elaborate this root node by listing the major quality attributes that the system is required to exhibit.
- Under each quality attribute, record a specific refinement of that QA.
 - For example, performance might be decomposed into “data latency” and “transaction throughput.” Or it might be decomposed into “user wait time” and “time to refresh web page.”
 - The refinements that you choose should be the ones that are relevant to your system. Under each refinement, record the appropriate ASRs (usually expressed as QA scenarios).

Utility Tree

- Some ASRs might express more than one quality attribute and so might appear in more than one place in the tree.
 - That is not necessarily a problem, but it could be an indication that the ASR tries to cover too much diverse territory.
 - Such ASRs may be split into constituents that each attach to smaller concerns.
- Once the ASRs are recorded and placed in the tree, you can now evaluate them against the two criteria we listed above: the business value of the candidate ASR and the architectural impact of including it.
 - You can use any scale you like, but we find that a simple “H” (high), “M” (medium), and “L” (low) suffice for each criterion.

Utility Tree

- For **business value**,
 - High designates a **must-have** requirement,
 - Medium is for a requirement that is **important** but would **not lead** to project **failure** were it **omitted**.
 - Low describes a **nice requirement** to have but **not something worth** much **effort**.
- For **architectural impact**,
 - High means that **meeting** this **ASR** will profoundly **affect** the **architecture**.
 - Medium means that **meeting** this **ASR** will **somewhat affect** the **architecture**.
 - Low means that **meeting** this candidate **ASR** will have **little effect** on the architecture.

Utility Tree Example

- Table 16.5 shows a portion of a sample utility tree drawn from a health care application called Nightingale.
- Each ASR is labeled with a pair of “H,” “M,” and “L” values indicating:
 - a) the ASR’s business value and
 - b) its effect on the architecture.

Utility

TABLE 16.5 Tabular Form of the Utility Tree for the Nightingale ATAM Exercise

Quality Attribute	Attribute Refinement	ASR
Performance	Transaction response time	A user updates a patient’s account in response to a change-of-address notification while the system is under peak load, and the transaction completes in less than 0.75 second. (H,M)
		A user updates a patient’s account in response to a change-of-address notification while the system is under double the peak load, and the transaction completes in less than 4 seconds. (L,M)
Usability	Throughput	At peak load, the system is able to complete 150 normalized transactions per second. (M,M)
	Proficiency training	A new hire with two or more years’ experience in the business becomes proficient in Nightingale’s core functions in less than 1 week. (M,L)
		A user in a particular context asks for help, and the system provides help for that context, within 3 seconds. (H,M)
Configurability	Normal operations	A hospital payment officer initiates a payment plan for a patient while interacting with that patient and completes the process without the system introducing delays. (M,M)
		A hospital increases the fee for a particular service. The configuration team makes the change in 1 working day; no source code needs to change. (H,L)
Maintainability	Routine changes	A maintainer encounters search- and response-time deficiencies, fixes the bug, and distributes the bug fix with no more than 3 person-days of effort. (H,M)
		A reporting requirement requires a change to the report-generating metadata. Change is made in 4 person-hours of effort. (M,L)
	Upgrades to commercial components	The database vendor releases a new version that must be installed in less than 3 person-weeks. (H,M)

Key:
H=high
M=medium
L=low

Utility Tree

- Once you have a **utility tree** filled out, you can **use** it **to** make important **checks**. For instance:
 - A **QA** or **QA refinement without** any **ASR** is **not** necessarily an **error** or **omission** that needs to be rectified, **but** it is an **indication** that **attention** should be paid to **finding** out for sure **if there are unrecorded ASRs** in that area.
 - **ASRs** that **rate** a **(H,H)** rating are **obviously** the ones that **deserve** the **most attention** from you; these are the **most significant** of the significant requirements. A very large number of these might be a cause for concern about whether the system is achievable.
 - **Stakeholders** can **review** the **utility tree** to **make sure** their **concerns** are **addressed**. (An alternative to the organization we have described here is to use stakeholder roles rather than quality attributes as the organizing rule under “Utility.”)

Utility Tree (Summary)

- A way to record ASRs all in one place.
- Establishes priority of each ASR in terms of
 - Impact on architecture
 - Business or mission value
- ASRs are captured as scenarios.
- Root of tree is placeholder node called “Utility”.
- Second level of tree contains broad QA categories.
- Third level of tree refines those categories.

Tying the Methods Together

- How should you employ requirements documents, stakeholder interviews, Quality Attribute Workshops, PALM, and utility trees in concert with each other?
 - If you have a requirements process that gathers, identifies, and prioritizes ASRs, then use that and consider yourself lucky.
 - Otherwise use one or more of the other approaches.
 - If nobody has captured the business goals behind the system you're building, use PALM.
 - If important stakeholders have been overlooked in the requirements-gathering process, use interviews or a QAW.
 - Build a utility tree to capture ASRs along with their prioritization.
 - You can blend all the methods together
 - Use PALM as a “subroutine call” from a QAW for the business goal step
 - Use a quality attribute utility tree as a repository for the scenarios produced by a QAW.
- Pick the approach that fills in the biggest gap in your existing requirements.

Summary

- Architectures are driven by architecturally significant requirements: requirements that will have profound effects on the architecture.
 - Architecturally significant requirements may be captured from requirements documents, by interviewing stakeholders, or by conducting a Quality Attribute Workshop.
- Be mindful of the business goals of the organization.
 - Business goals can be expressed in a common, structured form and represented as scenarios.
 - Business goals may be elicited and documented using a structured facilitation method called PALM.
- A useful representation of quality attribute requirements is in a utility tree.
 - The utility tree helps to capture these requirements in a structured form.
 - Scenarios are prioritized.
 - This prioritized set defines your “marching orders” as an architect.